Review 2 — Presenter's Guide

Explainable Multilingual Fake News Detection using Contrastive Learning and Fact Verification for Low-Resource Languages · Batch 15 · 16 slides

> **How to use this guide.** Each slide has three parts: **WHAT IS ON THE SLIDE** (so you can point at it), **WHAT TO SAY** (a script you can speak almost word-for-word), and **WHY** (the reasoning, so you can answer follow-ups). Green boxes are things to say out loud. Red boxes are traps to avoid. At the end there is a list of likely reviewer questions with answers.

Part 0 — Vocabulary you must be comfortable with

Learn these eight terms first. Every technical question in your review will use at least one.

Term	Plain-English meaning
Embedding	A list of numbers that represents a piece of text. Ours has 768 numbers. Texts with similar meaning get similar lists. This is how a computer "reads".
Transformer	The type of neural network used by modern language models. It reads all words at once and works out how each word relates to every other word.
XLM-RoBERTa	A specific transformer that was pre-trained on 100 languages, including Tamil. We use it to turn an article into an embedding.
Frozen model	We use XLM-RoBERTa exactly as downloaded and do not change its internal weights. It only extracts features; it is not trained by us.
Classifier	A small neural network that takes the embedding and outputs a decision: Fake or Real.
Training / Validation / Test	Three separate parts of the data. Train = the model learns from it. Validation = we use it to choose settings. Test = locked away, used once at the very end to report an honest final score.
Low-resource language	A language with very little labelled data available for machine learning. Tamil is one; English is not.
Macro-F1	Our main score. It averages performance on the "fake" class and the "real" class equally, so the model cannot cheat by only being good at the common class.

Slide 1 — Title Slide

What is on the slide

Project title, the three student names with roll numbers, department and college, course code 22CDP72, batch number, project guide, and the date.

What to say (about 30 seconds)

> **Say:** "Good morning. Our project is *Explainable Multilingual Fake News Detection using Contrastive Learning and Fact Verification for Low-Resource Languages*. I am <name>, with <names>, from Computer Science and Design, guided by Dr. R. Thangarajan. For Review 2 we will present the six modules we have implemented and the results we measured."

Why the title is worded this way

- **Explainable** — the final system will show *why* it made a decision, not just the label.
- **Multilingual** — one model handles Tamil and English together, not two separate models.
- **Contrastive Learning** — the technique that makes the Tamil part work despite little data.
- **Fact Verification** — checking a claim against evidence, which is a later module.
- **Low-Resource Languages** — the research problem. Tamil has very little labelled fake-news data.

Slide 2 — Introduction

What is on the slide

Six bullet points: the problem, why Tamil is hard, what we built, the model we use, the technique we use, and the result we got.

What to say (about 90 seconds)

> **Say:** "Fake news spreads faster than anyone can check it by hand. Almost all research on detecting it is done in English. Tamil has very little labelled data — in our own corpus Tamil is only 5,408 records out of 69,008, which is 7.8 per cent. We have built a bilingual corpus of 69,008 labelled news articles from four public sources. We use XLM-RoBERTa, which places Tamil and English text into the same 768-number space, so one single model can handle both languages. On top of that we apply supervised contrastive learning, which lets a Tamil article learn from the much larger pool of English articles that carry the same label. The classifier we trained on those representations reaches 0.9763 macro-F1 on validation, and Tamil is within 0.009 of English."

Why each point is there

- **"Spreads faster than it can be verified"** — motivates automation. Human fact-checkers cannot scale.
- **"7.8 per cent"** — this single number is the heart of the project. Say it clearly. It is why we need contrastive learning rather than just training a Tamil model.
- **"One shared 768-dimensional space"** — this is the key idea. Both languages are described using the same 768 numbers, so knowledge learned from English is usable for Tamil.
- **"Within 0.009 of English"** — this proves the approach worked. Normally a language with 7.8 % of the data performs far worse.

Slide 3 — Objectives

What is on the slide

Six objectives, in the same order as the modules, so the reviewer can map objective → module.

What to say (about 60 seconds)

> **Say:** "Our objectives follow the pipeline. First, build a labelled Tamil–English corpus where every record's source is recorded and the splits are checked for leakage. Second, represent both languages in one shared space using XLM-RoBERTa. Third, strengthen that representation with supervised contrastive learning so the minority language benefits from the majority. Fourth, classify each article as fake or real. Fifth, evaluate with accuracy, precision, recall, macro-F1 and ROC-AUC, reported separately for Tamil and English. Sixth, extend the framework with fact verification and explainability in the remaining modules."

Why these particular objectives

- **"Recorded source provenance"** — we know which of the four sources every article came from. This matters because a model could otherwise cheat by learning the writing style of one source.
- **"Leakage-controlled splits"** — see Slide 11. If the same article appeared in both training and test, the score would be falsely high.
- **"Reported separately for Tamil and English"** — an overall score can hide poor Tamil performance, because English is 92 % of the data. Reporting per language is honest.

Slides 4, 5 and 6 — Literature Survey

What is on the slides

Three tables covering 15 recent papers (2025–2026). Each row gives the paper title, author and date, the model that paper proposed, and the future work that paper itself recommends.

What to say (about 2 minutes total — do not read every row)

> **Say:** "We surveyed fifteen recent papers. Rather than read each one, let me show you the pattern we found. Papers 1, 5, 7 and 11 use advanced deep-learning models — graph networks, GANs, transfer learning — but all of them work in a single high-resource language. Paper 2, Ni'mah and colleagues, is our base paper: it uses contrastive learning for low-resource fake news, and it is the closest to our work. Papers 6, 8 and 12 look at low-resource languages such as Bangla and Amharic, which shows the problem is recognised but not solved for Tamil. Now look at the last column, Future Work. Paper 1 says 'extend to multiple languages'. Paper 2 says 'add multilingual support'. Papers 4, 7, 9, 10 and 13 all say 'add explainable AI'. In other words, the authors of these papers themselves identify multilingual support and explainability as the open problems. Our project addresses exactly those two gaps together."

Why this framing is strong

> You are not claiming the papers are bad. You are using their **own stated future work** as evidence that your research gap is real. This is much harder to argue against than saying "nobody has done this."

The three or four papers you must be able to discuss

No.	Paper	What it did, and how it relates to us
2	Ni'mah et al., *A simple contrastive embedding framework for low-resource fake news detection*, Neural Computing and Applications, 2025	**This is our base paper.** It uses contrastive learning plus an ensemble of MLPs for Indonesian fake news, where the data is extremely imbalanced (1 real to 24 fake). Its own future work says "add multilingual support". We extend it from one language to two, and from Indonesian to a Tamil–English setting.
1	Ji & Lv, *Explainable multi-granularity attribution reasoning framework*, 2026	Focuses on explainability — showing which parts of an article drove the decision. Its future work asks for multiple languages. We combine explainability with multilingual support.
6	Hossen et al., *BiLSTM-LIME*, 2026	Combines a BiLSTM classifier with LIME explanations. Its future work explicitly says "improve detection for low-resource languages" — which is our target.
12	Chalehchaleh et al., *Addressing data*	Tackles the same scarcity problem, but by generating synthetic data

scarcity in multilingual fake news detection, 2025	with a large language model. We solve it differently — by letting the minority language learn from the majority through contrastive learning, with no synthetic data.

Do not read all fifteen rows aloud. You will lose the room and run out of time. Point at the table, give the pattern, and name three or four papers.

Slide 7 — Research Gap

What is on the slide

Six gaps, moving from the general problem to the specific opening for our work.

What to say (about 75 seconds)

> **Say:** "We identified six gaps. One: research concentrates on high-resource languages; Tamil has very little labelled data and almost no published baselines. Two: the low-resource data that does exist is skewed by construction — fact-checking desks mainly publish claims that are already suspected to be false, so the data is not a fair sample of real news. Three: translating Tamil into English before detection loses linguistic nuance and adds translation errors. Four: most models output only a label; they never check the claim against external evidence. Five: existing systems are black boxes — they do not show which part of the article caused the decision. Six: no single framework combines all four capabilities: multilingual support, contrastive learning, fact verification and explainability."

Why each gap matters — the reasoning behind the words

- **Gap 2, "skewed by construction"**, is the most sophisticated point on this slide. A fact-checking website only writes about claims someone already doubted. So a dataset built from it contains far more fake than real. A model can then score well simply by guessing "fake" every time. This is exactly why we report **macro-F1** and a **majority-class baseline** later.
- **Gap 3, translation**, justifies our choice of a multilingual model. The alternative approach — translate Tamil to English, then use an English detector — introduces errors twice: once in the translation, once in the detector.
- **Gaps 4 and 5** are the reason our project continues past Module 6 into fact verification and explainability.

Slide 8 — Problem Statement

What is on the slide

Four statements: the gap, the data constraint, what we build, and what is finished.

What to say (about 45 seconds)

> **Say:** "The problem in one sentence: there is no unified framework that detects fake news accurately in both Tamil and English while staying evidence-grounded and explainable. The constraint is that Tamil forms only 7.8 per cent of the available data, so a model trained on Tamil alone has too little supervision to generalise. So this project builds a Tamil–English framework that puts both languages in one XLM-RoBERTa space, applies supervised contrastive learning so the minority language benefits from the majority, and classifies each article as

> fake or real. Modules 1 to 6 are implemented and evaluated."

> **The logical chain to keep in your head:** Tamil has little data → training Tamil alone will not work → so put Tamil and English in one shared space → then use contrastive learning to connect them → so Tamil borrows learning from English. If you can say that chain, you understand the project.

Slide 9 — Proposed Work: List of Modules

What is on the slide

Ten modules in two columns. The left column, in solid navy, is Modules 1–6, which are complete. The right column, outlined, is Modules 7–10, which remain. A green arrow and a hand-off card explain that the classifier's output feeds into fact verification.

What to say (about 90 seconds)

> **Say:** "This is the full pipeline. The filled boxes on the left are done; the outlined boxes on the right are the remaining work. Module 1 collects the data. Module 2 cleans it and splits it. Module 3 converts text into numbers using XLM-RoBERTa. Module 4 improves those numbers with contrastive learning. Module 5 classifies fake or real. Module 6 evaluates the result. That is the six modules we present today — 60 per cent of the pipeline. The output then hands off to Module 7, fact verification, which retrieves evidence for the claim; Module 8 fuses the classifier and the verifier into one decision; Module 9 explains the decision at word level; and Module 10 is the web demo."

Why the modules are in this order

1. **Data before models.** You cannot extract features from data you have not cleaned. Every later module depends on Module 2's output being correct.
2. **Features before learning.** Module 3 turns text into 768 numbers. Module 4 cannot improve a representation that does not exist yet.
3. **Representation before classification.** The classifier in Module 5 never sees raw text — it only sees the numbers produced by Modules 3 and 4. This separation is deliberate: it means we can improve the representation without retraining the classifier from scratch.
4. **Evaluation last.** Module 6 measures everything before we build on top of it.

> **If asked "why is fact verification not done yet?"** — Say: "It depends on a working classifier and a clean claim corpus. Building it before Module 6 was evaluated would mean building on an unverified foundation. That is why the pipeline is ordered this way."

Slide 10 — Module Completion Status

What is on the slide

A six-row table — each module, what was completed, and a "Completed" status — followed by the overall progress statement and the four verification checks.

What to say (about 60 seconds)

> **Say:** "All six modules are complete. Module 1 produced 69,089 records from four public sources with provenance retained. Module 2 produced 69,008 clean records with a stratified 70-15-15 split. Module 3 produced 138,016 embeddings. Module 4 trained the contrastive model and recorded an eight-configuration ablation. Module 5 trained the classifier and selected its configuration on validation. Module 6 evaluated with accuracy, precision, recall, macro-F1 and ROC-AUC, per language. That is six of the ten planned modules — 60 per cent. And these are verified, not claimed: the leakage check passed, all six data-quality checks passed, all eight ablation configurations are recorded, and the test split has not been used."

Why the four verification lines are on the slide

Anyone can write "Completed" in a table. These four lines are the **evidence** that the work was done properly. Mentioning that the **test split has not been used** tells an experienced reviewer that you understand experimental discipline. It is the single most credible sentence on the slide.

Slide 11 — Modules 1 and 2: Dataset and Preprocessing

What is on the slide

Left: the four data sources with real/fake counts. Right: the six preprocessing steps, the final split table, and the leakage check result.

Module 1 — Dataset Collection

Source	Language	Real	Fake	Total
WELFake	English	28,882	34,790	63,672
aanirudh_headlines	Tamil	2,285	2,893	5,178
Daily Thanthi	Tamil	214	11	225
BOOM Tamil	Tamil	2	12	14
Total	—	**31,383**	**37,706**	**69,089**

> **Say:** "We collected 69,089 records from four public sources. WELFake supplies the English side. For Tamil we use a published research dataset of Tamil fake-news headlines, plus Daily Thanthi as a mainstream newspaper and BOOM Tamil as a fact-checking desk. Note the imbalance across languages: 63,673 English against 5,416 Tamil. That imbalance is the research problem, not a mistake in our collection."

Why four sources and not one

- **WELFake** is a large, well-known English benchmark, so our English results are comparable to other work.
- **aanirudh_headlines** is a peer-reviewed Tamil corpus that ships real fake/real labels — it supplies 5,178 of our 5,416 Tamil records.
- **Daily Thanthi** is a mainstream newspaper, used as a source of genuine Tamil news.
- **BOOM Tamil** is a fact-checking desk, so its content skews towards debunked claims.

> **Honest limitation to state if asked:** Daily Thanthi and BOOM Tamil together are only 239 records, and their labels come from a source-type assumption rather than per-article annotation. The other 5,178 Tamil records carry real published labels. Say this plainly if a reviewer probes label quality — it is 4.4 % of the Tamil data and we have documented it.

Module 2 — Data Preprocessing, step by step

Step	What happened	Why it is necessary
Missing-value handling	514 missing titles repaired, 0 remaining	A missing title would produce an empty or misleading text field, and the model would learn from noise.
Short-document removal	70 documents removed (English under 5 words, Tamil under 2)	A two-word English fragment carries no evidence either way. Different thresholds are used because Tamil is a highly agglutinative language — one Tamil word carries the meaning of several English words.
Duplicate removal	11 exact duplicates removed: 69,019 → 69,008	If a duplicate landed in training and its twin in testing, the model would appear to generalise when it had simply memorised.
Label encoding	0 = real, 1 = fake; 0 null, 0 invalid	Machine-learning models need numbers, not words. Fixing fake = 1 also fixes the meaning of "precision" and "recall" later.
Language identification	Verified against the declared field — 0 mismatches	Per-language results are only meaningful if the language tag is correct.
Stratified split	70 / 15 / 15, seed 42, stratified on (language, label)	See below — the most important step on this slide.

What "stratified split" means, and why it matters here

A plain random split would take 15 % of rows at random for testing. Because Tamil is only 7.8 % of the data, a random draw could easily leave the test set with very few Tamil records — or with almost no Tamil *real* records, since those are rarer still. Then the Tamil score would be meaningless.

Stratifying on (language, label) means we form four groups — English-real, English-fake, Tamil-real, Tamil-fake — and take exactly 70 / 15 / 15 from *each group separately*. Every group is represented in every split in the same proportion as in the full corpus.

Split	Records	Share
Train	48,305	70 %
Validation	10,351	15 %
Test	10,352	15 %

What the leakage check is

After splitting, we tested whether any record appears in more than one split. The result was **train ∩ validation = 0, train ∩ test = 0, validation ∩ test = 0**. Zero overlap.

> **Say:** "Leakage is when the same article appears in both training and testing. The model then recognises it rather than generalising, and the score looks far better than the model really is. We checked all three pairs and found zero overlap."

> **The number to remember:** seed 42. This means the split is reproducible — anyone re-running our code gets exactly the same 48,305 / 10,351 / 10,352 rows.

Slide 12 — Modules 3 and 4: Features and Contrastive Learning

Module 3 — Feature Extraction with XLM-RoBERTa

The four-step chain on the slide: `Tamil / English text → SentencePiece tokenizer → xlm-roberta-base encoder → mean pooling → L2 normalisation → 768-d unit vector`.

Step 1: What is a tokenizer, and what is SentencePiece?

A neural network cannot read letters. A **tokenizer** chops text into small pieces called tokens and gives each a number. **SentencePiece** is the particular tokenizer XLM-RoBERTa uses; it has a vocabulary of about 250,000 pieces covering 100 languages.

> **Why this matters for Tamil:** Tamil script is included in that vocabulary directly. We do **not** transliterate Tamil into English letters and we do **not** translate it. The Tamil text goes in as Tamil. This is exactly the problem identified in Research Gap 3.

Step 2: Why XLM-RoBERTa specifically?

- **It is multilingual by design.** It was pre-trained on 100 languages at once, so Tamil and English already sit in one shared space. Two separate monolingual models could not do this — their number spaces would be unrelated.
- **Tamil is one of the 100 languages.** Many multilingual models exclude it.
- **We use it frozen.** We do not modify its weights. It is a fixed feature extractor. This was a deliberate scope decision: fine-tuning a 270-million-parameter model needs far more GPU time than a student project has, and the frozen features already work well.

> **Say if asked "why not fine-tune it?":** "Fine-tuning the full encoder was outside our compute budget, and it was not necessary — the frozen features already reach 0.9763 macro-F1. We instead put our training effort into the representation-learning stage, Module 4."

Step 3: What is mean pooling, and why not CLS?

XLM-RoBERTa outputs *one vector per token* — so a 300-word article gives roughly 300 vectors. We need **one** vector for the whole article. Two standard ways to do this:

- **CLS pooling** — take only the special first token's vector, and treat it as a summary.
- **Mean pooling** — average all the token vectors. We weight the average by the attention mask so that padding (empty filler added to make all texts the same length) contributes nothing.

> **We did not guess — we measured.** Our ablation tested both. Mean pooling beat CLS pooling in **all four** matched comparisons (0.9763 vs 0.9713; 0.9734 vs 0.9686; 0.9683 vs 0.9661; 0.9736 vs 0.9594). That is why the final model uses mean pooling. This is a strong answer if a reviewer asks how you chose.

Step 4: What is L2 normalisation and why do it?

Each article is now 768 numbers. **L2 normalisation** rescales those numbers so the vector has length exactly 1 — mathematically, it places every article on the surface of a sphere.

Why:

- **Length should not matter, direction should.** Without normalising, a long article produces a "bigger" vector than a short one purely because of length. After normalising, only the *direction* of the vector — which encodes meaning — matters.
- **The contrastive loss needs it.** Contrastive learning compares texts using cosine similarity, which is defined between directions. On unit vectors, cosine similarity is simply the dot product, which is fast and stable.
- **It is what the classifier expects.** The classifier was trained on unit vectors, so anything fed to it later must be normalised the same way.

Result of Module 3: 138,016 embedding vectors — that is 69,008 articles × 2 pooling strategies (mean and CLS), each 768 numbers, stored as float32, with max sequence length 512.

Module 4 — Supervised Contrastive Learning (SupCon)

The idea in one sentence: pull articles with the same label closer together in the 768-number space, and push articles with different labels further apart.

How positive and negative pairs are formed

- Training happens in **batches** of 256 articles.
- For any one article (the "anchor"), every other article in the batch **with the same label** is a **positive**. Every article with the **opposite label** is a **negative**.
- Crucially, a positive can be in a **different language**. A Tamil fake-news article and an English fake-news article in the same batch are a positive pair.

> **Say — this is the core contribution, say it slowly:** "Because positives are defined by label and not by language, a Tamil article is pulled towards English articles that share its label. That is the mechanism by which Tamil, which is only 7.8 per cent of the data, borrows learning from the English majority. We measured 130.1 positive pairs per anchor in the first epoch, with zero anchors left without a positive."

Why contrastive learning rather than just training a classifier

- A plain classifier learns a **boundary**. With very little Tamil data, the boundary in the Tamil region of the space is poorly defined.
- Contrastive learning first reshapes the **space itself** so that fake articles cluster together and real articles cluster together — *across both languages*. A boundary drawn afterwards is then much easier to place correctly.
- This is precisely the technique our base paper (Ni'mah et al.) used for low-resource Indonesian. Our contribution is extending it to a multilingual setting.

The training settings, and what each one does

Setting	Value	What it does
Loss	Supervised contrastive	The mathematical objective being minimised.
Temperature	0.07	Controls how sharply the model separates similar from dissimilar. A small value makes it focus hard on the most confusable pairs.
Optimizer	AdamW	The algorithm that adjusts the weights. AdamW adapts the step size per weight and handles weight decay correctly.
Learning rate	5×10^{-4}	How big each adjustment step is. Too large and training is unstable; too small and it never converges.
Batch size	256	How many articles are compared at once. Larger batches give more positive and negative pairs per step, which contrastive learning benefits from.
Max epochs / actually run	30 / 13	An epoch is one pass over the training data. Training stopped at 13 because of early stopping.

> **What "early stopping" means:** after each epoch we check the score on the validation set. If it has not improved for 5 consecutive epochs, we stop and keep the best model. This prevents **overfitting** — the model memorising the training data instead of learning general patterns.

Slide 13 — Modules 5 and 6: Classification and Evaluation

Module 5 — The MLP classifier

MLP stands for Multi-Layer Perceptron — the simplest kind of neural network. Ours is `768 → 512 → 128 → 2`. It takes the 768-number embedding and reduces it in stages down to 2 numbers: one score for "real", one for "fake".

Component	Value	What it does and why
Loss function	CrossEntropyLoss	The standard loss for classification. It penalises the model more heavily the more confidently it is wrong.
Optimizer	AdamW	Same reasoning as Module 4 — reliable, adaptive, and the standard choice for this kind of network.
Learning rate	1×10^{-3}	Larger than Module 4's because this network is much smaller and simpler, so it can take bigger steps safely.
Batch size	512	Larger than Module 4's because this stage does not need many pairs per batch — it only classifies.
Epochs / early stopping	50 / patience 8	Stops when validation macro-F1 stops improving for 8 epochs.
ReLU	—	The activation function between layers. Without it the three layers would collapse mathematically into one, and the network could only learn straight-line boundaries.
Dropout 0.1	—	During training, randomly switches off 10 % of the connections each step. This forces the network not to depend on any single feature, which reduces overfitting.

Module 6 — Every metric on the slide, explained

Metric	What it measures	Our value
Accuracy	Out of all articles, what fraction did we label correctly. *Weakness:* if 90 % of data were fake, guessing "fake" always would score 0.90 and be useless.	0.9765
Precision (macro)	Of the articles we called fake, what fraction really were fake. High precision = few false alarms.	0.9769
Recall (macro)	Of the articles that really were fake, what fraction did we catch. High recall = few misses.	0.9758
Macro F1	The harmonic mean of precision and recall, averaged over both classes *equally*. This is	**0.9763**

	our headline metric because it cannot be inflated by the majority class.	
ROC-AUC	The probability that the model gives a randomly chosen fake article a higher fake-score than a randomly chosen real one. 0.5 = random guessing, 1.0 = perfect.	0.9969

How to read a confusion matrix (there are two on the slide)

A confusion matrix is a 2×2 table. Rows are what the article *actually* is; columns are what the model *predicted*. The diagonal is correct; off-diagonal is wrong.

	Predicted Real	Predicted Fake
Actually Real	4,544 — correct	152 — false alarm
Actually Fake	91 — missed fake	5,564 — correct

> **Say:** "Out of 10,351 validation articles we made 243 errors in total: 152 real articles wrongly flagged as fake, and 91 fake articles missed. For the Tamil subset of 811 articles, 10 false alarms and 16 misses. In this application the missed fakes are the costlier error, because a fake article that slips through is the one that does harm."

The per-language table — the most important result you have

Language	n	Accuracy	Macro F1	ROC-AUC
English	9,540	0.9773	0.9770	0.9971
Tamil	811	0.9679	0.9678	0.9951

> **Say — this is your headline claim:** "Tamil macro-F1 is 0.9678 against English 0.9770. The gap is only 0.0092, even though Tamil is just 7.8 per cent of the corpus. Normally a language with that little data performs far worse. This gap is the evidence that the cross-language contrastive objective is doing real work."

> **Critical — say this and do not let it be discovered instead:** every number on this slide is a **validation** result, not a test result. The test split has deliberately not been used yet. **Say:** "All figures here are from the validation split. We are holding the test split back and will evaluate it once, at the end, so that our final reported number is unbiased." A reviewer who spots this on their own will think you were hiding it. A reviewer who hears it from you will think you are careful.

Slide 14 — 60 % Demo: Ablation and Best Result

What an ablation study is

An **ablation study** means changing one design decision at a time and re-measuring, so you can prove *which* choices actually mattered. Without it, you could claim "our model works" but not say why.

We tested **2 pooling strategies × 2 sampling strategies × 2 methods = 8 configurations**:

- **Pooling:** mean vs CLS (explained on Slide 12).
- **Sampling:** *uniform* — draw training batches at random; *language-aware* — deliberately force more Tamil into each batch.
- **Method:** *CE-MLP* — the cross-entropy classifier; *SupCon* — the contrastive model with a linear head.

What the results show

Configuration	Macro F1	Reading
mean · uniform · CE-MLP **(selected)**	**0.9763**	Best. This is our final model.
mean · language-aware · SupCon	0.9736	Second. SupCon's best result.
mean · language-aware · CE-MLP	0.9734	Forcing more Tamil per batch did not help CE-MLP.
cls · uniform · CE-MLP	0.9713	Same as the winner but CLS pooling — lower.
cls · language-aware · CE-MLP	0.9686	
mean · uniform · SupCon	0.9683	
cls · uniform · SupCon	0.9661	
cls · language-aware · SupCon	0.9594	Weakest configuration.

> **Say:** "Three conclusions. First, mean pooling beat CLS pooling in every single matched comparison — that is four out of four. Second, the best overall configuration was mean pooling with uniform sampling and the cross-entropy MLP, at 0.9763. Third — and we report this openly — the contrastive model reached 0.9683, slightly below the cross-entropy classifier here. We record every configuration including the weaker ones, because selection was made on validation macro-F1 alone."

> **Be ready for: "So contrastive learning did not help?"** Answer honestly and completely: "At this data scale, with frozen embeddings, the cross-entropy classifier was slightly ahead — 0.9763 against 0.9683. But two things. First, both are far above the strongest classical baseline at 0.8107, so the transformer representation is doing the heavy lifting. Second, the contrastive setup is what keeps Tamil within 0.009 of English despite having 7.8 per

cent of the data. We report the honest ranking rather than only the configuration that flatters us."

The comparison table on Slide 13 — why the baselines are there

Model	Macro F1	Why it is in the table
Majority-class baseline	0.3533	Always predicts the more common class. This is the floor — any real model must beat it. It proves our task is not trivially easy.
Logistic regression (CLS)	0.7893	A simple classical model on the same embeddings. Shows how much the neural classifier adds.
Logistic regression (mean)	0.8107	The strongest classical baseline.
SupCon + linear head	0.9683	Our contrastive model.
MLP classifier (selected)	**0.9763**	Our final model. **+0.1656** over the strongest classical baseline.

Slide 15 — References

What to say (15 seconds)

Say: "Nine references in IEEE format. Reference 2, Ni'mah and colleagues in Neural Computing and Applications, is our base paper. References 7, 8 and 9 are the methods we directly use: XLM-RoBERTa, supervised contrastive learning, and the Tamil dataset."

Know reference [2] cold — it is your base paper and you will be asked about it. Ni'mah et al., "A simple contrastive embedding framework for low-resource fake news detection," *Neural Computing and Applications*, vol. 37, pp. 21407–21433, 2025.

Slide 16 — Thank You

Say: "That completes Modules 1 to 6, which is 60 per cent of the pipeline. Thank you — we are happy to take questions."

Part 2 — Likely reviewer questions, with answers

Q1. Why XLM-RoBERTa and not BERT or mBERT?

BERT is English-only, so it cannot handle Tamil at all. mBERT is multilingual but XLM-RoBERTa was trained on far more data across 100 languages and generally performs better on low-resource languages. Critically, both Tamil and English are covered by its SentencePiece vocabulary, so we never translate or transliterate.

Q2. Your accuracy is 97 %. Is that not suspiciously high?

Three honest points. First, WELFake is a curated benchmark where fake and real articles come from systematically different sources, so the task is easier than detecting fake news in the wild. Second, we ran a leakage check and found zero overlap between splits, so the score is not inflated by duplicates. Third, this is a validation figure — our test split is untouched, and the final number will be reported from it once.

Q3. Why is Tamil almost as good as English with only 7.8 % of the data?

Because the two languages share one embedding space, and because contrastive positives are defined by label rather than by language. A Tamil fake article is pulled towards English fake articles. So Tamil effectively borrows supervision from the English majority instead of relying on its own 5,408 records.

Q4. Why report macro-F1 instead of accuracy?

Our corpus is imbalanced — 37,704 fake against 31,304 real, and far more English than Tamil. Accuracy can be high while one class performs badly. Macro-F1 averages the two classes equally, so it exposes that. Our majority-class baseline makes the point: 0.5463 accuracy but only 0.3533 macro-F1.

Q5. Why have you not used the test set?

Because every setting we chose — pooling, sampler, method, number of epochs — was chosen using validation scores. If we also reported on the test set now, and later changed anything, the test score would no longer be an unbiased estimate. Standard practice is to evaluate the test split once, at the end. That is what we will do.

Q6. What exactly is left to do?

Four modules. Module 7, fact verification: retrieve evidence for a claim and check whether the evidence supports or refutes it. Module 8, decision fusion: combine the classifier's output with the verifier's. Module 9, explainability: show which words drove the prediction. Module 10, an application demo.

Q7. How is your work different from the base paper?

The base paper, Ni'mah et al., applies contrastive learning to a single low-resource language, Indonesian, and its own stated future work asks for multilingual support. We extend the idea to a genuinely bilingual setting — Tamil and English in one shared space, with cross-language positive pairs — and our pipeline continues into fact verification and explainability, which that paper does not address.

Q8. Why is the split 70 / 15 / 15?

It is a standard split that leaves enough data to train on while keeping validation and test sets large enough to give stable estimates. Our validation and test sets are 10,351 and 10,352 records, which is comfortably large. The split is stratified on language and label and fixed with seed 42, so it is reproducible.

Q9. What does "frozen" mean, and does it limit you?

Frozen means we do not update XLM-RoBERTa's internal weights; we only use it to produce embeddings. It limits peak performance in principle, because a fine-tuned encoder could adapt to our specific data. In practice the frozen features already reach 0.9763 macro-F1, and fine-tuning a 270-million-parameter model was outside our compute budget. It is a documented scope decision, not an oversight.

Q10. How do you know your Tamil labels are correct?

5,178 of the 5,416 Tamil records come from a published, peer-reviewed dataset that ships real fake/real labels. The remaining 239 come from Daily Thanthi and BOOM Tamil, where the label follows from the source type. We have documented that limitation rather than hiding it — it affects 4.4 % of the Tamil data.

Part 3 — Numbers to memorise

Number	What it is
69,089 → 69,008	Records collected, then records after cleaning.
5,408 Tamil / 63,600 English	Language split after cleaning — Tamil is 7.8 %.
48,305 / 10,351 / 10,352	Train / validation / test.
768	Numbers in each embedding.
138,016	Embeddings generated (69,008 × 2 pooling strategies).
0.9763	Best validation macro-F1 — the headline result.
0.9770 vs 0.9678	English vs Tamil macro-F1 — a gap of only 0.0092.
0.3533	Majority-class baseline macro-F1 — the floor.
+0.1656	Our gain over the strongest classical baseline.
8	Ablation configurations, all recorded.
6 of 10 = 60 %	Modules complete.

The three sentences that will most impress a reviewer

1."Tamil macro-F1 is within 0.009 of English despite Tamil being only 7.8 per cent of the corpus."

2."All figures are validation results; the test split is held out and will be evaluated once."

3."We record every ablation configuration, including the ones where our contrastive model came second."

Each one signals that you understand your own experiment rather than only its best number.